# Graphs of cube root functions

**1** Consider the function $y = \sqrt[3]{x}$.

**a** Complete the table of values. Round any values to two decimal places if necessary.

| $x$ | $-100$ | $-10$ | $-8$ | $-3$ | $-1$ | $0$ | $1$ | $3$ | $8$ | $10$ | $100$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | | | | | |

**b** Sketch the graph of $y = \sqrt[3]{x}$.

**c** Is $y = \sqrt[3]{x}$ an increasing function or a decreasing function?

**d** Is there any restriction on the value of $x$?

**e** State the range of the function.

**f** Does $y = \sqrt[3]{x}$ have a limiting value?

**g** Does $y = \sqrt[3]{x}$ have an asymptote?

**h** As $x$ gets larger and larger, what value does $y$ approach?

**i** For $x \geq 0$, describe the rate of increase of the function as $x$ increases.

**j** For $x \geq 0$, describe the rate of describe the rate of increase of the function as $x$ increases.

**2** Consider the function $y = -\sqrt[3]{x}$.

**a** Complete the table of values. Round any values to two decimal places if necessary.

| $x$ | $-100$ | $-10$ | $-8$ | $-3$ | $-1$ | $0$ | $1$ | $3$ | $8$ | $10$ | $100$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | | | | | |

**b** Sketch the graph of the function.

**c** Can the function values ever be positive?

**d** Can the function value ever be $0$?

**e** State the domain of the function.

**f** State the range of the function.

**g** Is $y = -\sqrt[3]{x}$ an increasing function or a decreasing function?

**h** Describe the rate of increase/decrease of the function $y = -\sqrt[3]{x}$ on either side of $x = 0$.

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**3** Consider the function $y = \sqrt[3]{-x}$.

**a** Complete the table of values. Round any values to two decimal places if necessary.

| $x$ | $-100$ | $-10$ | $-8$ | $-3$ | $-1$ | $0$ | $1$ | $3$ | $8$ | $10$ | $100$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| $y$ | | | | | | | | | | | |

**b** Sketch the graph of $y = \sqrt[3]{-x}$.

**c** Is $y = \sqrt[3]{-x}$ an increasing function or a decreasing function?

**d** State the domain of the function.

**e** State the range of the function.

**f** As $x$ approaches $-\infty$, what does $y$ approach?

**g** As $x$ approaches $\infty$, what does $y$ approach?

**4** Consider the function $y = \sqrt[3]{x} + 3$.

**a** Can $y$ ever be negative?

**b** As $x$ gets larger and larger, what value does $y$ approach?

**c** Determine the $y$-intercept of the curve.

**d** How many $x$-intercepts does it have?

**e** Sketch the graph of $y = \sqrt[3]{x} + 3$.

**5** Consider the function $y = -5\sqrt[3]{x}$.

**a** State the domain of the function.

**b** State the range of the function.

**c** Sketch the graph of $y = -5\sqrt[3]{x}$.

**6** Consider the function $f(x) = \sqrt[3]{x-1}$.

**a** Complete following table of values:

| $x$ | $-7$ | $0$ | $1$ | $2$ | $9$ |
|---|---|---|---|---|---|
| $\sqrt[3]{x-1}$ | $-2$ | | | | |

**b** Sketch the graph of the function.

**c** State the domain of the function.

**d** State the range of the function.

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**7** Consider the function $f(x) = \sqrt[3]{x} - 1$.

**a** Complete following table of values:

| $x$ | $-8$ | $-1$ | $0$ | $1$ | $8$ |
|---|---|---|---|---|---|
| $\sqrt[3]{x} - 1$ | $-3$ | | | | |

**b** Sketch the graph of the function.

**c** State the domain of the function.

**d** State the range of the function.

**8** Consider the following functions:

**i** Sketch the graph of the function.

**ii** State the domain of the function.

**iii** State the range of the function.

**a** $f(x) = \frac{1}{2}\sqrt[3]{x}$

**b** $f(x) = \frac{\sqrt[3]{-x}}{3}$

**c** $f(x) = \sqrt[3]{\frac{1}{3}x}$

**d** $f(x) = \sqrt[3]{x-1} + 4$

**e** $f(x) = \sqrt[3]{-x} - 5$

**f** $f(x) = \sqrt[3]{\frac{x+4}{2}}$

**g** $f(x) = \sqrt[3]{-(x-5)}$

**h** $f(x) = 3\sqrt[3]{x} - 2$

**i** $f(x) = \frac{\sqrt[3]{4x}}{2}$

**j** $f(x) = \sqrt[3]{-\frac{x}{2}}$

## Characteristics of cube root functions

**9** Describe how the function $y = 5\sqrt[3]{x}$ differs from $y = \sqrt[3]{x}$.

**10** State the domain and range of $y = \sqrt[3]{x} - 4$.

**11** Consider the function $y = \sqrt[3]{x-4}$.

**a** State the domain of the function.

**b** State the range of the function.

**c** Compare the rate of increase of $y = \sqrt[3]{x-4}$ and $y = \sqrt[3]{x}$.

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